

DFIR Case Study: USB-Delivered Fileless Malware & Memory Forensics

Digital Forensics & Incident Response Report

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1. Agency / Company Introduction

CyberTrace Forensics Lab is a digital forensics and incident response agency specializing in memory forensics, malware analysis, and incident response. The agency follows industry standard frameworks such as NIST 800-86 to ensure evidence integrity and court-defensible reporting.

2. Investigation Team Members:

Member	Responsibilities
Jawad Iqbal	Volatility Analyst
Saad Rizwan	First Responder
Shahzaib Mumtaz	Documentation / Expert Witness
Ali Raza	Autopsy Analyst

1 - Team Members Names and Responsibilities

3. Abstract of the Project

This project investigates a **fileless malware attack** initiated through a **malicious USB device**. The attacker used a disguised VBS shortcut to trigger a multi-stage script chain (**VBS → BAT → Python**), leading to shellcode injection and a reverse shell connection. A Wazuh agent installed on the victim's PC detected the unauthorized USB insertion, triggering a security alert and initiating the forensic investigation. Memory forensics using Volatility revealed the full attack chain, attacker IP, and evidence of data exfiltration. Autopsy confirmed malicious USB contents and attacker-created artifacts.

4. Purpose of this report

The purpose of this report is to:

- A. Document the forensic investigation process
- B. Reconstruct the attack chain
- C. Identify malicious activity in memory and disk artifacts
- D. Present findings in a structured, examiner-ready format
- E. Provide recommendations to prevent similar attacks

5. Case Brief / Introduction and Background of the Project

A Honda employee inserted a USB device into his office PC. The organization's **Wazuh agent**, which monitors endpoint activity, immediately detected the insertion of an external USB device. Since USB usage was prohibited under company policy, Wazuh flagged this as a **high-severity alert**. This alert triggered the initiation of a digital forensic investigation. Upon inspection, the USB contained a fake folder named **"Toyota New Models"**, which was actually a malicious VBS shortcut. When executed, it silently launched:

- a) A hidden VBS script

- b) A BAT file
- c) A malicious Python scripts

The Python script downloaded shellcode, allocated RWX memory, injected the payload, and established a reverse shell to the attacker. A memory dump was acquired for forensic analysis.

6. Scope and Objectives

a) Scope

- 1) Memory forensics
- 2) Disk artifact analysis
- 3) Wazuh alert correlation
- 4) Script chain reconstruction
- 5) Network connection analysis
- 6) Identification of attacker activity

b) Objectives

- 1) Identify initial infection vector
- 2) Reconstruct execution chain
- 3) Detect shellcode injection
- 4) Identify attacker IP
- 5) Confirm data exfiltration
- 6) Correlate Wazuh alert with forensic timeline
- 7) Provide recommendations

7. Disclaimer

This investigation is conducted strictly for academic and research purposes.

8. Chain of Custody

Case Number: *DF-LAB/2025- USB- 001*

Offense: *Unauthorized Access / Malware Execution via USB*

Victim: *University Lab Environment (Training Scenario)*

Suspect: *Unknown (Simulated Attacker)*

Location of Seizure: *Digital Forensics Lab – Workstation Area*

Date/Time Seized: *11-December- 2025, 02:00 PM*

Description of Evidence

Item #	Description of Evidence	Serial/ID	Condition	Quantity
1	Seized Computer System (Victim PC)	PC-LAB-001	Good	1

2	USB Flash Drive (Attack Vector)	USB- ATTACK- 001	Minor scratches	1
3	VMDK Disk Image (Forensic Copy)	IMG- VMDK- 001	New/Good	1
4	Memory Dump (memdump.mem)	MEM- DUMP- 001	New/Good	1
5	Malicious Folder “Toyota Car Models”	MAL- FOLDER- 001	N/A	1 Folder
6	Exfiltration ZIP File (totoya.zip)	ZIP- EXFIL- 001	N/A	1 File
7	Wazuh USB Detection Logs	LOG- WAZUH- 001	New/Good	Multiple pages

Chain of Custody Record

Item #	Date/Time	Released by (Name/ID/Signature)	Received by (Name/ID/Signature)	Location/Comments
1	11-Dec- 2025 / 02:40 PM	Saad (First Responder)	Jawad (Volatility Analyst)	Seized PC handed over for forensic processing
2	11-Dec- 2025 / 02:40 PM	Saad (First Responder)	Ali (Autopsy Analyst)	USB device seized and transferred for disk imaging
3	11-Dec- 2025 / 02:45 PM	Saad (First Responder)	Ali (Autopsy Analyst)	VMDK image acquired from seized PC and given for Autopsy analysis
4	11-Dec- 2025 / 02:45 PM	Saad (First Responder)	Jawad (Volatility Analyst)	Memory dump acquired and transferred for volatile analysis
5	12-Dec- 2025 / 04:00PM	Ali (Autopsy Analyst)	Shahzaib (Documenter / Expert Witness)	Malicious folder extracted and documented
6	12-Dec- 2025 / 04:00 PM	Ali (Autopsy Analyst)	Shahzaib (Documenter / Expert Witness)	Exfiltration ZIP recovered and logged
7	12-Dec- 2025 / 04:05 PM	Jawad (Volatility Analyst)	Shahzaib (Documenter / Expert Witness)	Wazuh logs collected and added to evidence file

9. Search & Seizure Form

SEIZURE MEMO

Case Number: *DF- LAB/2025- USB- 001*

Item: *Victim Computer System (Seized for Forensic Examination)*

Date of Seizure: *11-December- 2025*

Time: *02:30 PM*

Location: *Digital Forensics Laboratory – Workstation Area*

Details of Person (Seizing Officer): *Saad Rizwan – First Responder*

Details of Person from whom item(s) seized: *Training Scenario – No real suspect involved*

Address / Telephone Number / Email: *University Lab Records*

Description of Item(s)

Description of item seized: *Desktop Computer (Victim Workstation)*

Make/Model: *Generic Lab Workstation (Training Environment)*

Serial Number: *PC- LAB- 001*

Colour: *Black*

Condition: *Good; system powered on at time of seizure*

Number of pages (if documents): *N/A*

Any other identifying marks: *University Lab Asset Tag #LAB- PC- 01*

Signatures

Name	Signature
Shahzaib Mumtaz	<i>Shahzaib mumtaz</i>
Saad Rizwan	<i>Saad Rizwan</i>

10. Methodology & Investigation Plan

a) Digital Forensics Investigation Process

- 1) Identification
- 2) Collection
- 3) Preservation
- 4) Analysis
- 5) Reporting

b) Wazuh Alert Correlation

- 1) Wazuh detected unauthorized USB insertion
- 2) Alert timestamp was used to correlate with Volatility session timestamps
- 3) This validated the start of the attack timeline

c) Lab Environment

- 1) Windows 10 victim machine

- 2) Kali Linux attacker machine
- 3) Volatility Workbench
- 4) Volatility 3
- 5) Autopsy
- 6) FTK Imager
- 7) Wazuh agent logs

11. Forensic Tools & Equipment's:

Tool	Version	Purpose
FTK Imager	4.7.3.81	Memory acquisition
Autopsy	4.22.1	Disk artifact analysis
Volatility Workbench	3.0	Memory forensics
Volatility 3	2.26.2	Advanced memory analysis
Wazuh	4.12	Endpoint monitoring & alerting
Python	3.13	Script analysis

2 - Tools and Technology Stack used in Case

12. Brief about Evidentiary Items

Item No	Description	Location
01	Memory Dump (memdump.mem)	Acquired from victim PC
02	USB Contents	VBS, BAT, Python files
03	Attacker-created ZIP file	toyota.zip
04	Wazuh Alert Logs	Endpoint security logs

3 - Brief Details of Acquired Evidences

13. Forensic Laboratory Evidence Labels:

- a) **E01 -- Seized Computer System**
Physical desktop PC collected from the incident location.
- b) **2. E02 -- Seized USB Device**
Kingston DataTraveler 3.0 USB drive recovered during seizure.
- c) **E03 -- Disk Image (Forensic Copy)**
The Vmdk image used for analysis in autopsy.
- d) **E04 -- Memory Image (Forensic Copy)**
RAM dump acquired from the seized PC for volatile analysis.
- e) **E05 -- Malicious Folder "Toyota Car Models"**

Suspicious folder located on the Desktop containing the malicious VBS, BAT, Python payload, shortcut file, and supporting components.

f) E06 -- Exfiltration Archive (toyota.zip)

ZIP archive recovered from the Documents folder.

g) E07 -- USB Detection Log

Wazuh alert showing recognition of an external USB device.

14. Specification of each [Evidentiary item]

a) E01 -- Seized Computer System

- 1) **Type:** Physical evidence
- 2) **Description:** Desktop PC collected from the victim environment.
- 3) **Relevance:** Primary host where the malicious activity occurred.

b) E02 -- Seized USB Device

- 1) **Type:** Physical evidence
- 2) **Description:** Kingston DataTraveler 3.0 USB drive.
- 3) **Relevance:** Identified as the initial infection vector used to deliver the malicious folder.

c) E03 -- Disk Image (Forensic Copy)

- 1) **Type:** Digital forensic evidence
- 2) **Description:** VMDK disk image acquired from the seized computer system and loaded directly into Autopsy for examination.
- 3) **Relevance:** Contains all disk-based artifacts, including the malicious “Toyota Car Models” folder, Prefetch execution traces, attacker activity, and the exfiltration archive (fcty.zip). This image served as the primary source for file system and timeline analysis.

d) E04 -- Memory Image (Forensic Copy)

- 1) **Type:** Digital forensic evidence
- 2) **Description:** RAM dump acquired for volatile analysis.
- 3) **Relevance:** Used to identify running processes, script execution, the attacker’s IP address and in memory payload activity.

e) E05 -- Malicious Folder “Toyota Car Models”

- 1) **Type:** Digital evidence
- 2) **Description:** Folder containing the staged attack chain: VBS launcher, BAT script, Python payload, shortcut file, and supporting files.
- 3) **Relevance:** Represents the complete malicious delivery package introduced via USB.

f) E06 -- Exfiltration Archive (toyota.zip)

- 1) **Type:** Digital evidence
- 2) **Description:** ZIP archive created during attacker activity.
- 3) **Relevance:** Indicates attacker activity for exfiltration.

g) E07 -- USB Detection Log

- 1) **Type:** Log evidence

- 2) **Description:** Wazuh alert showing detection of an external USB device.
- 3) **Relevance:** Establishes the initial point of compromise and confirms removable media-based attack.

15. Methods & Procedures for Analysis:

a) Memory Analysis

- 1) Volatility Workbench
- 2) Volatility 3

b) Disk Analysis

- 1) Autopsy
- 2) File extraction
- 3) Timeline analysis

c) Wazuh Log Analysis

- 1) USB insertion alert
- 2) Policy violation detection
- 3) Correlation with Volatility timestamps

16. Imaging Process

a) Disk Image Acquisition (VMDK)

Performed by: Saad (First Responder)

Tool/Environment: VMware Workstation

Process:

1. The system was already virtualized in VMware.
2. The existing **VMDK file** was identified in the VM directory.
3. The VMDK was copied to the forensic workstation as the **forensic disk image**.
4. Integrity hashes (MD5/SHA1) were generated and recorded.
5. Image was handed over to **Ali (Autopsy Analyst)** for file system analysis.



Windows 10.scoreboard	16/12/2025 22:01	SCOREBOARD File	8 KB
Windows 10.vmdk	16/12/2025 22:01	VMDK File	102,103,42...
Windows 10.vmsd	01/09/2025 22:11	VMware snapshot ...	0 KB
Windows 10.vmx	16/12/2025 22:01	VMware virtual m...	4 KB

1 - VMDK file located in VMware directory used as Forensic Disk Image

[-]	
Name	Windows 10.vmdk
Sector count	419430400
[-] MD5 Hash	
Computed hash	004819bd026af60891ba95ff14ccbb3d
[-] SHA1 Hash	
Computed hash	e2b0fde94879367d144b16d6752d0d1e9e175
[-] Bad Blocks List	
Bad block(s) in image	No bad blocks found in image

2 - MD5 and SHA1 hashes generated for VMDK

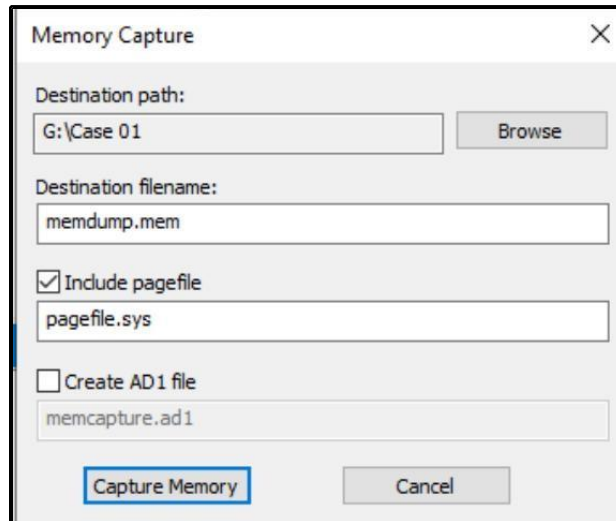
b) Memory Acquisition (RAM Image)

Performed by: Saad (First Responder)

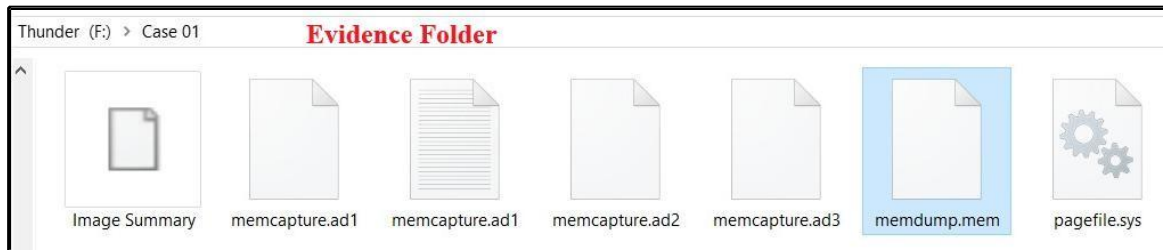
Tool Used: FTK Imager (Live Memory Capture)

Process:

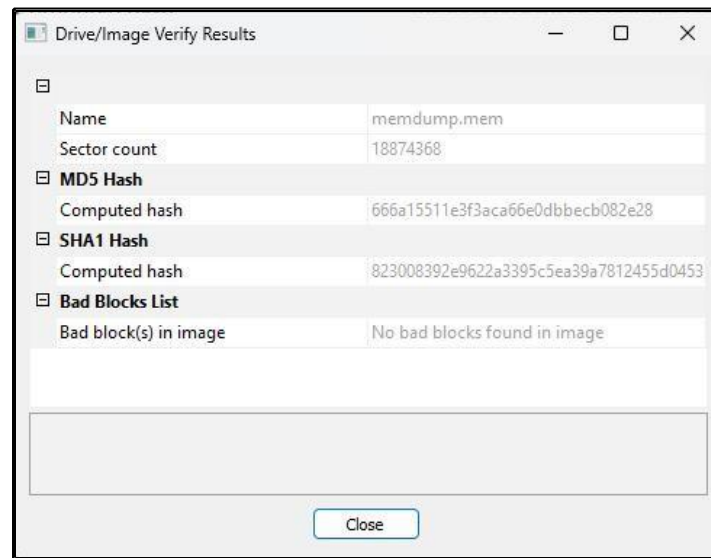
1. FTK Imager's **Memory Capture** module was executed inside the VM.
2. Full RAM was acquired and saved as memdump.mem.
3. The RAM image was exported to the host machine.
4. Integrity hashes were generated.
5. Memory dump was handed over to **Jawad (Volatility Analyst)**.



3 - FTK Imager memory capture dialog showing destination path and filename



4 - Evidence folder displaying memdump.mem, pagefile.sys, and associated files



5 - MD5 and SHA1 hashes generated for memdump.mem

17. Analysis:

a) Wazuh Alerts Analysis

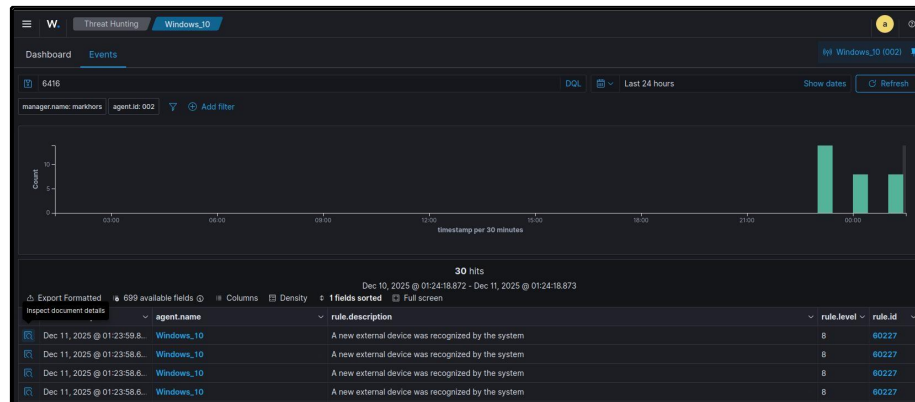
The investigation was initiated based on a security alert generated by the Wazuh monitoring system. The alert indicated that a **new external device was recognized by the system** on the victim machine running Windows 10.

1) Alert Summary:

- i. **Date/Time:** Dec 11, 2025 @ 01:23:59
- ii. **Agent Name:** Windows_10
- iii. **Rule Triggered:** A new external device was recognized by the system
- iv. **Event Type:** INFO
- v. **Channel:** Security
- vi. **Event Message:** "A new external device was recognized by the system."
- vii. **Agent IP:** 192.168.66.154
- viii. **Event ID:** 6416
- ix. **Rule ID:** 60227

2) Significance:

This alert served as the **starting point of the forensic investigation**, prompting the acquisition of memory and disk images for further analysis. It suggested possible USB-based malware execution or data exfiltration activity.



6 - Wazuh Dashboard – External Device Alert Timeline

Table	JSON
@timestamp	Dec 11, 2025 @ 01:23:59.871
_index	wazuh-alerts-4.x-2025.12.10
agent.id	002
agent.ip	192.168.66.164
agent.name	Windows_10
data.win.eventdata.classId	(eecd98-8888-425f-922a-dabf3de3f69a)
data.win.eventdata.className	WPD
data.win.eventdata.compatibleId	wpdusenum\fs SMD\Generic
data.win.eventdata.deviceDescription	Toyota Vehicles
data.win.eventdata.deviceId	SMD\WPDUSENUM\77_USBSTORWDisk&Ven_VendorCo&Prod_ProductCode&Rev_2_0085422291113622411220&8F53F56387-bdbf-11d0-94f2-00a0c91efbb)
data.win.eventdata.subjectDomainName	WORKGROUP
data.win.eventdata.subjectLogonId	0x3e7
data.win.eventdata.subjectUserName	DESKTOP-315152J
data.win.eventdata.subjectUserSid	S-1-5-18
data.win.system.channel	Security
data.win.system.computer	DESKTOP-315152J
data.win.system.eventID	6416
data.win.system.eventRecordID	16922
data.win.system.keywords	0x0200000000000000
data.win.system.level	0
data.win.system.message	"A new external device was recognized by the system."
	Subject: Security ID: S-1-5-18 Account Name: DESKTOP-315152J Account Domain: WORKGROUP Process ID: 0x3e7

7 - Wazuh Alert Log – Metadata of USB Recognition Event

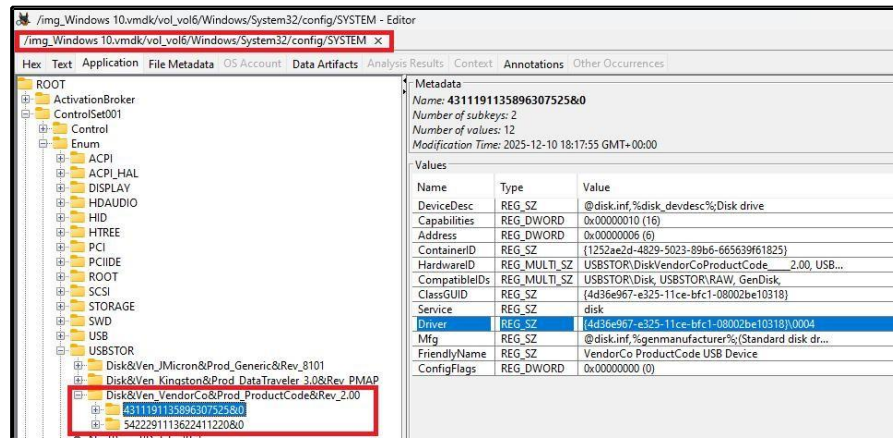
b) Autopsy Analysis

1) USB Device Connection

Event: A Kingston DataTraveler 3.0 USB device was connected to the system.

Evidence Source: *SYSTEM hive* → *Enum\USBSTOR*

Significance: Confirms that an external USB storage device was plugged into the machine. This device served as the delivery mechanism for the malicious files later found on the Desktop.



8 - USB Connection Verified from Registry

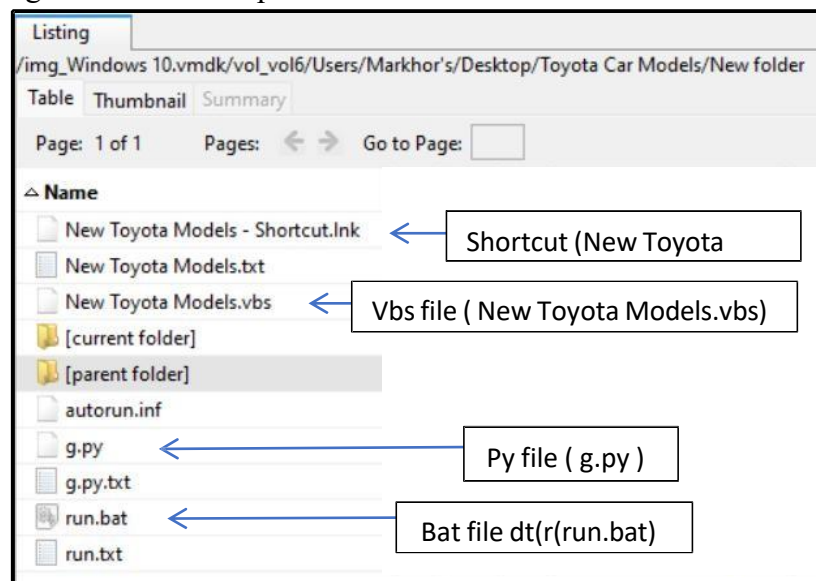
2) Malicious Files Appear on the Desktop

Event: A set of suspicious files was found in the user's Desktop directory, including:

New Toyota Models → *Shortcut.Ink* → *New Toyota Models.vbs* → *run.bat* → *g.py*

Evidence Source: File system view

Significance: These files indicate that the malicious payload was introduced through the USB device and staged on the Desktop for execution.



9 - Malicious Files that executed Malicious Payload

3) Execution of the VBS Script

Timestamp: 2025- 12- 11 01:38:39

Event: Creation of *WSCRIPT.EXE-*.pf*

Evidence Source: Prefetch

Significance: Confirms that the *.vbs* script was executed by the user. This marks the beginning of the script- based attack chain.

Name	S	C	O	Modified Time	Change Time	Access Time
WINHEX.EXE-70156968.pf				2025-09-25 21:27:56 PKT	2025-12-10 22:58:10 PKT	2025-12-11 00:46:36 PKT
WINRAR-X64-624.EXE-C94EB50D.pf				2025-09-23 19:29:52 PKT	2025-12-10 22:58:10 PKT	2025-12-11 00:46:36 PKT
WINRAR.EXE-BA8CD831.pf				2025-12-10 21:24:36 PKT	2025-12-10 22:58:10 PKT	2025-12-11 00:46:36 PKT
WINSAT.EXE-C345C80B.pf				2025-12-04 01:00:42 PKT	2025-12-10 22:58:10 PKT	2025-12-11 00:46:23 PKT
WINSTORE.APP.EXE-771FEC6D.pf				2025-12-02 11:24:33 PKT	2025-12-10 22:58:10 PKT	2025-12-11 00:46:36 PKT
WIRESHARK-4.4.9-X64.EXE-38093C2C.pf				2025-09-23 19:40:44 PKT	2025-12-10 22:58:10 PKT	2025-12-11 00:46:36 PKT
WLRMDR.EXE-A7C36FDD.pf				2025-12-10 23:01:03 PKT	2025-12-10 23:01:03 PKT	2025-12-11 00:46:23 PKT
WMIADAP.EXE-BB21CD77.pf				2025-12-10 23:02:20 PKT	2025-12-10 23:02:20 PKT	2025-12-11 00:46:23 PKT
WMIAPSRV.EXE-FC8436DD.pf				2025-12-10 21:36:41 PKT	2025-12-10 22:58:10 PKT	2025-12-11 00:46:23 PKT
WMIC.EXE-98223A30.pf				2025-12-11 02:14:51 PKT	2025-12-11 02:14:51 PKT	2025-12-11 02:14:51 PKT
WMIPRVSE.EXE-39F97B2D.pf				2025-12-04 09:13:21 PKT	2025-12-10 22:58:10 PKT	2025-12-11 00:46:23 PKT
WMIPRVSE.EXE-E8B8DD29.pf				2025-12-11 03:42:33 PKT	2025-12-11 03:42:33 PKT	2025-12-11 03:42:33 PKT
WMIPRVSE.EXE-E8B8DD29.pf				2025-12-11 00:39:08 PKT	2025-12-11 00:39:08 PKT	2025-12-11 00:39:08 PKT
WSCRIPT.EXE-3FF4D889.pf				2025-12-11 01:38:39 PKT	2025-12-11 01:38:39 PKT	2025-12-11 01:38:39 PKT

10 - Execution of VBS Script

4) Execution of the BAT File

Timestamp: Shortly after VBS execution

Event: Creation of *CMD.EXE-*.pf*

Evidence Source: Prefetch

Significance: Indicates that the *batch file (run.bat)* was executed as the second stage of the attack, it also indicates that the meterpreter session is started.

Metadata	
Name:	/img_Windows 10.vmdk/vol_vol6/Windows/Prefetch/CMD.EXE-0BD30981.pf
Type:	File System
MIME Type:	application/octet-stream
Size:	3232

Name	S	C	O	Modified Time	Change Time	Access Time
BOOTSTRAPPER.EXE-6925EBBC.pf				2025-12-02 12:59:53 PKT	2025-12-10 22:58:09 PKT	2025-12-11 00:46:36 PKT
BOOTSTRAPPER.EXE-BA3E275A.pf				2025-12-02 12:50:32 PKT	2025-12-10 22:58:09 PKT	2025-12-11 00:46:36 PKT
CERTUTIL.EXE-28F1E0C1.pf				2025-10-08 00:28:53 PKT	2025-12-10 22:58:09 PKT	2025-12-11 00:46:36 PKT
CHCP.COM-2CF9B15C.pf				2025-12-11 01:27:02 PKT	2025-12-11 01:27:02 PKT	2025-12-11 01:27:02 PKT
CHCP.COM-2CF9B15C.pf				2025-12-11 00:14:57 PKT	2025-12-11 00:14:57 PKT	2025-12-11 00:14:57 PKT
CMAKE.EXE-3241695E.pf				2025-12-04 00:09:11 PKT	2025-12-10 22:58:09 PKT	2025-12-11 00:46:22 PKT
CMD.EXE-0BD30981.pf				2025-12-11 02:11:21 PKT	2025-12-11 02:11:21 PKT	2025-12-11 02:11:21 PKT
CMD.EXE-0BD30981.pf				2025-12-11 00:34:15 PKT	2025-12-11 00:34:15 PKT	2025-12-11 00:34:15 PKT

11 - Indicator of Batch (bat) file was executed

5) Execution of the Python Script

Timestamp: 2025- 12- 11 01:38:43

Event: Creation of PYTHON.EXE-*.pf

Evidence Source: Prefetch

Significance: Confirms that the Python script (9.py / g.py) executed as the final stage of the malicious chain.

Listing						
/img_Windows 10.vmdk/vol_vol6/Windows/Prefetch						
Table Thumbnail Summary						
Page: 1 of 1 Pages: Go to Page:						
Name	S	C	O	Modified Time	Change Time	Access Time
OPENWITH.EXE-8B50D58B.pf				2025-12-04 11:41:40 PKT	2025-12-10 22:58:09 PKT	2025-12-11 00:46:36 PKT
Op-MSEdge.EXE-37D25F9A-00000001.pf				2025-09-09 23:47:19 PKT	2025-12-10 22:58:09 PKT	2025-12-11 00:46:36 PKT
PARTITIONWIZARD.EXE-169A369C.pf				2025-12-02 00:43:48 PKT	2025-12-10 22:58:09 PKT	2025-12-11 00:46:36 PKT
PCWRun.EXE-9C58AE8.pf				2025-12-10 22:46:27 PKT	2025-12-10 22:58:09 PKT	2025-12-11 00:46:22 PKT
PICKERHOST.EXE-DE4B8E61.pf				2025-11-25 22:12:26 PKT	2025-12-10 22:58:09 PKT	2025-12-11 00:46:36 PKT
PING.EXE-4A8A6853.pf				2025-12-11 01:46:01 PKT	2025-12-11 01:46:01 PKT	2025-12-11 01:46:01 PKT
PING.EXE-4A8A6853.pf				2025-12-11 00:20:40 PKT	2025-12-11 00:20:40 PKT	2025-12-11 00:20:40 PKT
POWERSHELL.EXE-CA1AE517.pf				2025-12-11 02:11:35 PKT	2025-12-11 02:11:35 PKT	2025-12-11 02:11:35 PKT
POWERSHELL.EXE-CA1AE517.pf				2025-12-11 00:33:48 PKT	2025-12-11 00:33:48 PKT	2025-12-11 00:33:48 PKT
PRINTUI.EXE-05AF7FAC.pf				2025-12-11 01:27:21 PKT	2025-12-11 01:27:21 PKT	2025-12-11 01:27:21 PKT
PRINTUI.EXE-05AF7FAC.pf				2025-12-11 00:15:14 PKT	2025-12-11 00:15:14 PKT	2025-12-11 00:15:14 PKT
PRODISCOVERBASIC.EXE-7B3B0BCC.pf				2025-09-25 21:26:53 PKT	2025-12-10 22:58:09 PKT	2025-12-11 00:46:36 PKT
PRODISCOVERRELEASE61100BASIC.-454D15D6.pf				2025-09-23 19:36:43 PKT	2025-12-10 22:58:09 PKT	2025-12-11 00:46:36 PKT
PW-FREE-ONLINE.TMP-AE810871.pf				2025-12-02 00:33:14 PKT	2025-12-10 22:58:09 PKT	2025-12-11 00:46:36 PKT
PW-FREE-ONLINE.TMP-E3219C2B.pf				2025-12-02 00:33:18 PKT	2025-12-10 22:58:09 PKT	2025-12-11 00:46:36 PKT
PWFREE-64BIT-ONLINE.TMP-D6E5A6CF.pf				2025-12-02 00:37:45 PKT	2025-12-10 22:58:09 PKT	2025-12-11 00:46:36 PKT
PY.EXE-FFC9A196.pf				2025-12-10 21:51:01 PKT	2025-12-10 22:58:09 PKT	2025-12-11 00:46:22 PKT
PYTHON-3.14.1-AMD64.EXE-CA2E7668.pf				2025-12-04 11:43:36 PKT	2025-12-10 22:58:09 PKT	2025-12-11 00:46:22 PKT
PYTHON-3.14.1-AMD64.EXE-F3168A61.pf				2025-12-04 11:43:34 PKT	2025-12-10 22:58:09 PKT	2025-12-11 00:46:36 PKT
PYTHON.EXE-D09EFF13.pf				2025-12-11 01:38:43 PKT	2025-12-11 01:38:43 PKT	2025-12-11 01:38:43 PKT

12 - Verification of Python Script Execution

6) Attacker-Driven System Commands

After the execution of the malicious scripts, Autopsy identified Prefetch files for several administrative utilities:

i. tasklist.exe Executed

Evidence: TASKLIST.EXE-*.pf

Meaning: Used to enumerate running processes.

Hex	Text	Application	File Metadata	OS Account	Data Artifacts	Analysis Results	Context	Annotations	Other Occurrences
Metadata									
Name:	/img_Windows 10.vmdk/vol_vol6/Windows/Prefetch/SCHTASKS.EXE-8B6144A9.pf								
Type:	File System								
MIME Type:	application/octet-stream								
Size:	3345								
File Name Allocation:	Allocated								
Metadata Allocation:	Allocated								

Listing						
/img_Windows 10.vmdk/vol_vol6/Windows/Prefetch						
Table Thumbnail Summary						
Page: 1 of 1 Pages: Go to Page:						
Name	S	C	O	Modified Time	Change Time	Access Time
SVCHOST.EXE-D0F686CF.pf				2025-12-11 02:30:50 PKT	2025-12-11 02:30:50 PKT	2025-12-11 02:30:50 PKT
SVCHOST.EXE-DAF72364.pf				2025-12-05 18:37:17 PKT	2025-12-10 22:58:09 PKT	2025-12-11 00:46:23 PKT
SVCHOST.EXE-DB0681F5.pf				2025-12-11 02:30:49 PKT	2025-12-11 02:30:49 PKT	2025-12-11 02:30:49 PKT
SVCHOST.EXE-DB425447.pf				2025-12-11 03:45:27 PKT	2025-12-11 03:45:27 PKT	2025-12-11 03:45:27 PKT
SVCHOST.EXE-DB425447.pf				2025-12-11 00:40:23 PKT	2025-12-11 00:40:23 PKT	2025-12-11 00:40:23 PKT
SVCHOST.EXE-DD4D71FD.pf				2025-12-10 23:00:47 PKT	2025-12-10 23:00:47 PKT	2025-12-11 00:46:23 PKT
SVCHOST.EXE-E23BD784.pf				2025-12-04 09:14:41 PKT	2025-12-10 22:58:09 PKT	2025-12-11 00:46:23 PKT
SVCHOST.EXE-ECB476FF.pf				2025-12-10 23:01:47 PKT	2025-12-10 23:01:47 PKT	2025-12-11 00:46:23 PKT
SVCHOST.EXE-F0617438.pf				2025-12-11 03:42:34 PKT	2025-12-11 03:42:34 PKT	2025-12-11 03:42:34 PKT
SVCHOST.EXE-F0617438.pf				2025-12-11 00:44:32 PKT	2025-12-11 00:44:32 PKT	2025-12-11 00:44:32 PKT
SVCHOST.EXE-F630A562.pf				2025-12-11 03:45:28 PKT	2025-12-11 03:45:28 PKT	2025-12-11 03:45:28 PKT
SVCHOST.EXE-F6604993.pf				2025-12-10 23:27:11 PKT	2025-12-10 22:58:09 PKT	2025-12-11 00:46:23 PKT
SVCHOST.EXE-FF256082.pf				2025-12-10 23:11:00 PKT	2025-12-10 23:11:00 PKT	2025-12-11 00:46:23 PKT
SVCINSTY64.EXE-CC6EACBE.pf				2025-12-11 01:27:24 PKT	2025-12-11 01:27:24 PKT	2025-12-11 01:27:24 PKT
SVCINSTY64.EXE-CC6EACBE.pf				2025-12-11 00:15:17 PKT	2025-12-11 00:15:17 PKT	2025-12-11 00:15:17 PKT
SVCTRL64.EXE-1BDDA055.pf				2025-12-04 09:14:26 PKT	2025-12-10 22:58:09 PKT	2025-12-11 00:46:23 PKT
SYSTEMINFO.EXE-3EAAF1C2.pf				2025-12-11 02:21:31 PKT	2025-12-11 02:21:31 PKT	2025-12-11 02:21:31 PKT
SYSTEMPROPERTIESCOMPUTERNAME-449B662F.pf				2025-12-02 13:15:39 PKT	2025-12-10 22:58:09 PKT	2025-12-11 00:46:23 PKT
SYSTEMSETTINGS.EXE-BE0858C5.pf				2025-12-08 21:39:16 PKT	2025-12-10 22:58:09 PKT	2025-12-11 00:46:23 PKT
TASKHOSTW.EXE-2E5D4B75.pf				2025-12-11 03:02:06 PKT	2025-12-11 03:02:06 PKT	2025-12-11 03:02:06 PKT
TASKHOSTW.EXE-2E5D4B75.pf				2025-12-10 23:26:45 PKT	2025-12-10 23:26:45 PKT	2025-12-11 00:46:22 PKT
TASKLIST.EXE-F58BCF08.pf				2025-12-11 02:21:37 PKT	2025-12-11 02:21:37 PKT	2025-12-11 02:21:37 PKT
TASKMGR.EXE-4C8500BA.pf				2025-12-02 12:23:16 PKT	2025-12-10 22:58:09 PKT	2025-12-11 00:46:23 PKT

13 - Verification of Enumeration of Running Processes

ii. schtasks.exe Executed

Evidence: SHTASKS.EXE-*.pf

Meaning: Used to create or manage scheduled tasks, suggesting an attempt to establish persistence.

Hex	Text	Application	File Metadata	OS Account	Data Artifacts	Analysis Results	Context	Annotations	Other Occurrences
Metadata									
Name:	/img_Windows 10.vmdk/vol_vol6/Windows/Prefetch/SHTASKS.EXE-8B6144A9.pf								
Type:	File System								
MIME Type:	application/octet-stream								
Size:	3345								

Listing						
/img_Windows 10.vmdk/vol_vol6/Windows/Prefetch						
Table Thumbnail Summary						
Page: 1 of 1 Pages: Go to Page:						
Name	S	C	O	Modified Time	Change Time	Access Time
RUNTIMEBROKER.EXE-94B34D1F.pf				2025-12-11 03:02:09 PKT	2025-12-11 03:02:09 PKT	2025-12-11 03:02:09 PKT
RUNTIMEBROKER.EXE-D72E00CE.pf				2025-12-10 23:00:47 PKT	2025-12-10 23:00:47 PKT	2025-12-11 00:46:22 PKT
RUNTIMEBROKER.EXE-F8127469.pf				2025-12-11 03:45:28 PKT	2025-12-11 03:45:28 PKT	2025-12-11 03:45:28 PKT
ReadyBoot				2025-09-02 10:18:15 PKT	2025-12-10 22:58:09 PKT	2025-12-10 22:58:51 PKT
ResPriHMSStaticDb.ebd				2025-12-02 12:16:32 PKT	2025-12-10 22:58:09 PKT	2025-12-11 00:47:20 PKT
SC.EXE-6C4D4413.pf				2025-12-11 02:13:15 PKT	2025-12-11 02:13:15 PKT	2025-12-11 02:13:15 PKT
SHTASKS.EXE-8B6144A9.pf				2025-12-11 02:21:36 PKT	2025-12-11 02:21:36 PKT	2025-12-11 02:21:36 PKT
SCREENCLIPPINGHOST.EXE-17C3B3F6.pf				2025-10-05 18:55:53 PKT	2025-12-10 22:58:09 PKT	2025-12-11 00:46:36 PKT

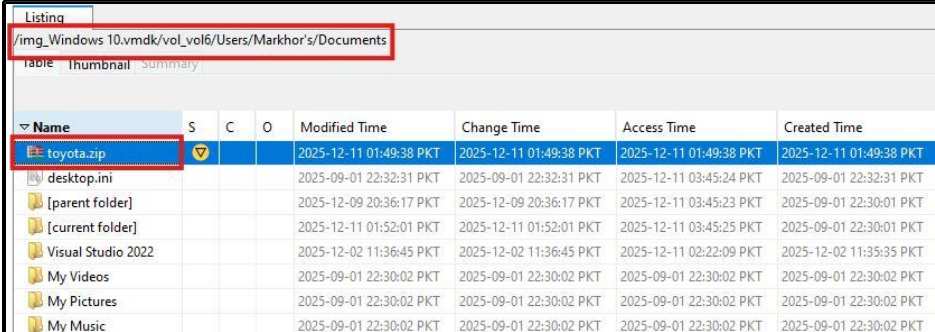
14 - Verification of Attempts to Establish Persistence

7) Creation of Exfiltration Archive

Event: Creation of fcty.zip in the user's Documents folder

Evidence Source: File timestamps

Significance: The ZIP archive appears to have been created to package data for exfiltration. Its timestamp is significantly later than the initial compromise, indicating it was generated during the attacker's activity window.



Name	S	C	O	Modified Time	Change Time	Access Time	Created Time
toyota.zip				2025-12-11 01:49:38 PKT	2025-12-11 01:49:38 PKT	2025-12-11 01:49:38 PKT	2025-12-11 01:49:38 PKT
desktop.ini				2025-09-01 22:32:31 PKT	2025-09-01 22:32:31 PKT	2025-12-11 03:45:24 PKT	2025-09-01 22:32:31 PKT
[parent folder]				2025-12-09 20:36:17 PKT	2025-12-09 20:36:17 PKT	2025-12-11 03:45:23 PKT	2025-09-01 22:30:01 PKT
[current folder]				2025-12-11 01:52:01 PKT	2025-12-11 01:52:01 PKT	2025-12-11 03:45:25 PKT	2025-09-01 22:30:01 PKT
Visual Studio 2022				2025-12-02 11:36:45 PKT	2025-12-02 11:36:45 PKT	2025-12-11 02:22:09 PKT	2025-12-02 11:35:35 PKT
My Videos				2025-09-01 22:30:02 PKT	2025-09-01 22:30:02 PKT	2025-09-01 22:30:02 PKT	2025-09-01 22:30:02 PKT
My Pictures				2025-09-01 22:30:02 PKT	2025-09-01 22:30:02 PKT	2025-09-01 22:30:02 PKT	2025-09-01 22:30:02 PKT
My Music				2025-09-01 22:30:02 PKT	2025-09-01 22:30:02 PKT	2025-09-01 22:30:02 PKT	2025-09-01 22:30:02 PKT

15 - Detection of Unknown and Suspicious ZIP File

8) Assessment

The Autopsy analysis provides clear disk- based evidence of a staged malware attack delivered through an external USB device. The presence of malicious scripts on the Desktop, combined with Prefetch files for *wscript.exe*, *cmd.exe*, and *python.exe*, confirms that the VBS → BAT → Python execution chain ran on the system. Additional Prefetch artefacts for *tasklist.exe*, *schtasks.exe* show that system commands were executed after the initial compromise, indicating active attacker interaction. The creation of the ZIP archive in the user's Documents folder further demonstrates that data was prepared for exfiltration. Overall, the disk artefacts collectively validate the full attack sequence, from initial USB insertion to script execution, attacker activity, and data staging.

c) Volatility Analysis

1) Overview

This analysis was conducted on a Windows memory image using the Volatility framework to examine suspected fileless malware activity. The objective was to determine how the attack began, how it executed in memory, and whether any evidence of command execution or data exfiltration was present.

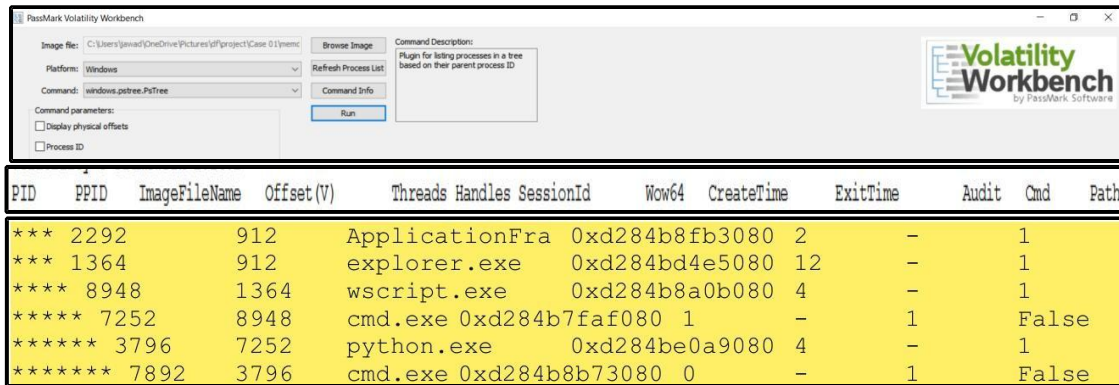
2) Process Execution Analysis

Using the *windows.pstree* plugin, the process hierarchy was examined to understand how the malicious activity began and progressed. The analysis showed that the attack originated from *explorer.exe*, indicating that the initial execution was triggered by user interaction. From there, a clear multi-stage execution chain was observed:

- *explorer.exe* launched *wscript.exe*, which executed a VBS script.

- **wscript.exe** then started **cmd.exe** to run a batch file
- **cmd.exe** finally launched **python.exe** (PID 3796), which acted as the main malicious payload

This sequence demonstrates a staged attack that gradually transitioned from legitimate system processes into malicious execution.



PID	PPID	ImageFileName	Offset(V)	Threads	Handles	SessionId	Wow64	CreateTime	ExitTime	Audit	Cmd	Path
***	2292		912	ApplicationFra	0xd284b8fb3080	2		-	-	1		
***	1364		912	explorer.exe	0xd284bd4e5080	12		-	-	1		
***	8948		1364	wscript.exe	0xd284b8a0b080	4		-	-	1		
*****	7252		8948	cmd.exe	0xd284b7faf080	1		-	1	False		
*****	3796		7252	python.exe	0xd284be0a9080	4		-	-	1		
*****	7892		3796	cmd.exe	0xd284b8b73080	0		-	1	False		

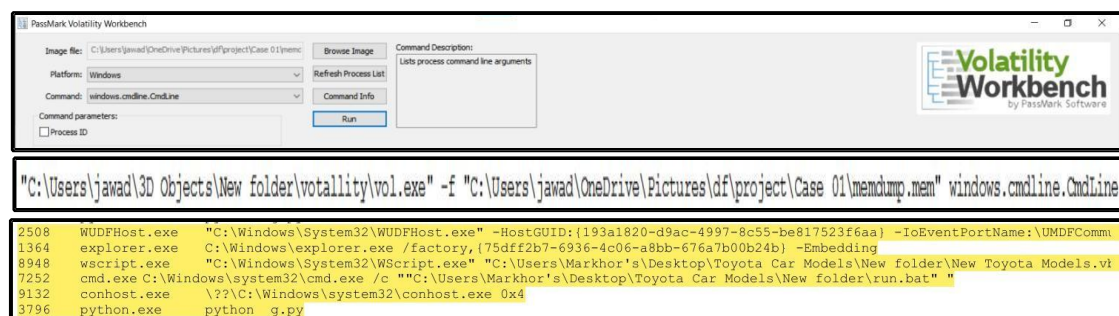
16 - Execution chain from explorer.exe to python.exe

3) Command-Line Activity

The **windows.cmdline** plugin was used to recover the exact commands executed by each process. The analysis revealed that:

- wscript.exe executed a script named “**New Toyota Models.vbs**” from the Desktop directory
- cmd.exe executed **run.bat** from the same location
- python.exe executed **g.py**, which contained the malicious logic

The use of a legitimate looking file name indicates **social engineering**, where the user was misled into opening a malicious script disguised as a harmless file.



PID	ImageFileName	CommandLine
2508	WUDFHost.exe	"C:\Windows\System32\WUDFHost.exe" -HostGUID:{193a1820-d9ac-4997-8c55-be817523f6aa} -IoEventPortName:\UMDFCommu
1364	explorer.exe	C:\Windows\explorer.exe /factory,(75dff2b7-6936-4c06-a8bb-676a7b00b24b) -Embedding
8948	wscript.exe	"C:\Windows\System32\WScript.exe" "C:\Users\Markhor's\Desktop\Toyota Car Models\New folder\New Toyota Models.vi
7252	cmd.exe	C:\Windows\system32\cmd.exe /c "C:\Users\Markhor's\Desktop\Toyota Car Models\New folder\run.bat"
9132	conhost.exe	??C:\Windows\system32\conhost.exe 0x4
3796	python.exe	python g.py

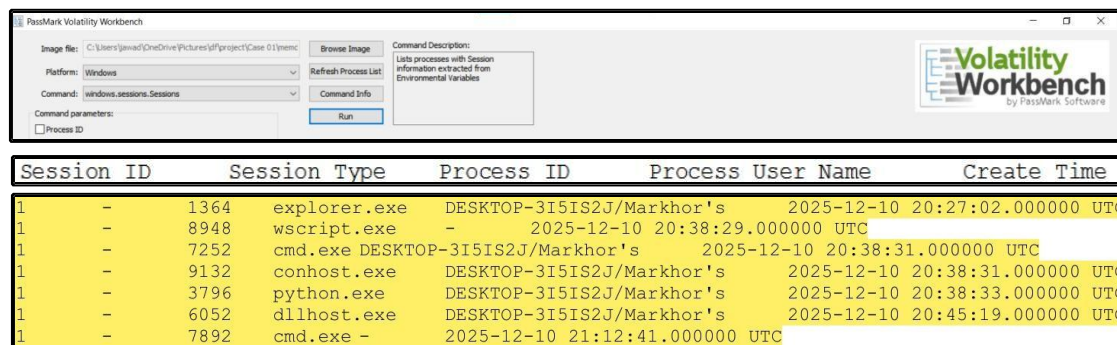
17 - windows.cmdline output showing command-line arguments for malicious processes.

4) Timeline and Session Context

To determine how quickly the attack occurred and under which user context, the windows.sessions plugin was analyzed. The analysis showed that the attack occurred under the user session “Markhor”, and all malicious processes were launched within a very short time window:

- wscript.exe at 20:38:29 UTC
- cmd.exe at 20:38:31 UTC
- python.exe at 20:38:33 UTC

The rapid execution of these stages indicates that the attack was automated and required minimal user interaction after the initial trigger.



Session ID	Session Type	Process ID	Process User Name	Create Time
1	-	1364	explorer.exe DESKTOP-3I5IS2J/Markhor's	2025-12-10 20:27:02.000000 UTC
1	-	8948	wscript.exe - 2025-12-10 20:38:29.000000 UTC	
1	-	7252	cmd.exe DESKTOP-3I5IS2J/Markhor's	2025-12-10 20:38:31.000000 UTC
1	-	9132	conhost.exe DESKTOP-3I5IS2J/Markhor's	2025-12-10 20:38:31.000000 UTC
1	-	3796	python.exe DESKTOP-3I5IS2J/Markhor's	2025-12-10 20:38:33.000000 UTC
1	-	6052	dllhost.exe DESKTOP-3I5IS2J/Markhor's	2025-12-10 20:45:19.000000 UTC
1	-	7892	cmd.exe - 2025-12-10 21:12:41.000000 UTC	

18 - Malicious activity under the Markhor user session

5) In-Memory Code Injection

The *windows.malfind* plugin was used to identify suspicious memory regions associated with injected code. The analysis revealed that *python.exe* contained memory regions with **read, write, and execute (RWX)** permissions and no backing file on disk. Disassembly of these regions showed patterns commonly associated with shellcode, including XOR-based decryption routines and position-independent code. This strongly indicates that malicious code was **injected directly into memory**, which is a common technique used by fileless malware to avoid detection.



PID	Process	Start VPN	End VPN	Tag	Protection	CommitCharge	PrivateMemory	File output	Notes	Hexdump	Disasm
3796	python.exe	0x4040000	0x4072fff		VadS	PAGE_EXECUTE_READWRITE	51	1	Disabled	N/A	
<pre> eb 27 5b 53 5f b0 65 fc ae 75 fd 57 59 53 5e 8a '[S_e..u.WYS^ 06 30 07 48 ff c7 48 ff c6 66 81 3f a7 c8 74 07 .0.H..H..f?.t. 80 3e 65 75 ea eb e6 ff e1 e8 d4 ff ff ff 11 0c .>eu..... 0f 31 4d 27 60 7d 07 85 83 07 07 07 07 0b 07 2e .1M'')..... 0x4040000: jmp 0x4040029 0x4040002: pop rbx 0x4040003: push rbx 0x4040004: pop rdi 0x4040005: mov al, 0x65 0x4040007: cld 0x4040008: scasb al, byte ptr [rdi] 0x4040009: jne 0x4040008 0x404000b: push rdi 0x404000c: pop rcx 0x404000d: push rbx 0x404000e: pop rsi 0x404000f: mov al, byte ptr [rsi] 0x4040011: xor byte ptr [rdi], al 0x4040013: inc rdi 0x4040016: inc rsi 0x4040019: cmp word ptr [rdi], 0xc8a7 0x404001e: je 0x4040027 0x4040020: cmp byte ptr [rsi], 0x65 0x4040023: jne 0x404000f 0x4040025: jmp 0x404000d 0x4040027: jmp rcx 0x4040029: call 0x4040002 0x404002e: adc dword ptr [rdi + rcx], ecx 0x4040031: xor dword ptr [rbp + 0x27], ecx </pre>											

19 - Highlighting RWX memory regions and injected shellcode within python.exe.

6) File Artifacts Observed in Memory

Although the attack was primarily fileless, the *windows.filescan* plugin identified a suspicious archive file:

`\Users\Markhor\Documents\toyota.zip`

This file has been created during the attacker's active session was used for exfiltration. The file name aligns with the original social engineering theme, suggesting continuity throughout the attack.

```

(kali@kali)-[~/Desktop/case 01]
$ python3 ~/volatility3/vol.py -f memdump.mem windows.filescan.FileScan | grep -i ".zip"

0xd284bc4163c0.0\Program Files\7-Zip\7-zip.dll
0xd284bc721c20 \Windows\System32\en-US\zipfldr.dll.mui
0xd284bce26a40 \Windows\SystemResources\zipfldr.dll.mun
0xd284bd346250 \Program Files\WindowsApps\Microsoft.OfficeHub.19.2512.33091.0_x64__8wekyb3d8bbwe\mini.zip.dll
0xd284bd34d140 \Windows\System32\en-US\zipfldr.dll.mui
0xd284bd351ab0 \Windows\SystemResources\zipfldr.dll.mun
0xd284bd54ddc0 \Users\Markhor's\Documents\toyota.zip
0xd284bd86d4e0 \Windows\System32\zipfldr.dll

```

20 - Identifying the suspicious archive file toyota.zip in the user's Documents directory

7) Network Activity

The *windows.netscan* plugin was used to identify network connections present in memory at the time of capture. The analysis revealed an established outbound connection from the victim system to an external host:

- Local IP: 192.168.66.164
- Remote IP: 192.168.66.154
- Port: 4444
- Associated process: python.exe (PID 3796)

Port 4444 is commonly used by Meterpreter payloads, strongly indicating that a **reverse shell** was successfully established.



The screenshot shows the Volatility Workbench interface. The 'Image file' is set to 'C:\Users\jared\OneDrive\Pictures\dfir\project\Case 01\mem'. The 'Platform' is 'Windows' and the 'Command' is 'windows.netscan.NetScan'. The 'Command parameters' are empty. The 'Run' button is visible. Below the interface, a table displays network connection data:

Offset	Proto	LocalAddr	LocalPort	ForeignAddr	ForeignPort	State	PID	Owner	Created
0xd284beaca8a0	TCPv4	192.168.66.164	54755	192.168.66.154	4444	ESTABLISHED	3796	python.exe	2025-12-10 20:38:47

21 - showing an established connection from python.exe on port 4444

8) Assessment

The evidence shows that the system was compromised using a fileless malware attack that started through social engineering. The attacker used multiple scripts to run a malicious payload directly in memory, gain remote access to the system, and prepare data for exfiltration. No permanent malware files were found on the disk, which shows why memory forensics was necessary to detect this attack.

18. Findings and Results

a) Analysis Findings

- 1) Wazuh detected unauthorized USB insertion
- 2) User executed malicious USB shortcut
- 3) Multi-stage script chain executed silently
- 4) Python script downloaded and injected shellcode
- 5) Reverse shell established to attacker
- 6) Attacker created ZIP file for exfiltration
- 7) No malware stored on disk in fileless attack

b) Results

Type	Value
Attacker IP	192.168.66.154
Malicious PIDs	8948, 7252, 3796
Malicious Files	New Toyota Models.vbs, run.bat, g.py
Exfil File	toyota.zip
Wazuh Alert	Unauthorized USB insertion

4 - Results after Analysis and Investigation

19. Conclusions and Recommendations

a) Conclusion

The investigation confirms a successful fileless malware attack delivered through social engineering. Wazuh played a critical role by detecting the unauthorized USB insertion and triggering the forensic investigation. Memory forensics revealed the full execution chain, shellcode injection, attacker IP, and exfiltration activity. Disk analysis supported the memory findings.

b) Recommendations

- 1) Enforce USB blocking policies
- 2) Strengthen Wazuh rules for endpoint monitoring
- 3) Block script execution (VBS/BAT/Python)
- 4) Deploy EDR solutions
- 5) Train employees
- 6) Monitor outbound traffic
- 7) Restrict PowerShell and Python usage

20. Limitations

During the course of this investigation, several limitations were encountered that affected the speed, efficiency, and correlation of the analysis:

a) Autopsy Performance and Loading Delays

Autopsy required a significant amount of time to load and process the VMDK image. This caused delays in browsing the file system, generating timelines, and extracting artifacts. Such performance issues are common when analyzing large virtual disk images or when running Autopsy on limited hardware.

b) Timestamp Correlation Across Different Tools

A major challenge was aligning timestamps from multiple forensic tools.

1. **Volatility** displayed timestamps in UTC.
2. **Autopsy** displayed timestamps in **local system time (PKT)**.
3. **Wazuh** logs used their own formatting and timezone context.

This required manual conversion and cross-checking to ensure accurate event correlation. The mismatch in time formats increased the complexity of building a unified timeline.

21. Annex / Appendices

a) Appendix A – Autopsy Generated Report

22. Glossary

a) Autopsy

A digital forensics tool used for analyzing disk images, recovering deleted files, examining timelines, and extracting artifacts such as browser history, registry entries, and user activity.

b) Volatility

A memory forensics framework used to analyze RAM dumps. It extracts processes, network connections, handles, DLLs, and malware artifacts from volatile memory.

c) Wazuh

A security monitoring and intrusion detection platform that generates alerts for suspicious activity such as USB insertions, file modifications, and system events.

d) VMDK (Virtual Machine Disk)

A virtual hard disk file used by VMware. It contains the full file system, partitions, and data of a virtual machine. In this investigation, it served as the forensic disk image.

e) Memory Dump (memdump.mem)

A captured snapshot of the system's RAM at a specific moment. Used to analyze running processes, malware execution, command history, and volatile artifacts.

f) FTK Imager

A forensic imaging tool used to acquire RAM and disk images. In this case, it was used to capture the memory dump from the running virtual machine.

g) External Device Recognition Alert

A Wazuh-generated event indicating that a USB or similar device was connected to the system. This alert served as the starting point of the investigation.

h) Timeline Analysis

A forensic technique that reconstructs events in chronological order using timestamps from logs, memory artifacts, and disk metadata.

i) First Responder

The individual responsible for seizing and preserving digital evidence at the scene. In this case, Saad performed the acquisition of memory and the VMDK.

j) Chain of Custody (CoC)

A documented record of who handled each piece of evidence, when it was transferred, and for what purpose. Ensures evidence integrity and admissibility.

k) MD5 / SHA1 Hash

Cryptographic hash values used to verify the integrity of forensic images. Matching hashes confirm that the evidence has not been altered.

l) UTC (Coordinated Universal Time)

A global time standard used by many forensic tools, including Volatility. Requires conversion to local time for correlation.

m) PKT (Pakistan Standard Time)

Local time zone used by the system and Autopsy. Required conversion when correlating with UTC-based tools.

n) **Malicious Folder (“Toyota Car Models”)**

A folder containing malicious scripts (VBS, BAT, Python) executed from the USB device.

o) **Exfiltration ZIP (toyota.zip)**

A ZIP file created by the malicious script to collect and compress data for exfiltration.

p) **Disk Image Analysis**

The process of examining the VMDK file to identify file system changes, artifacts, and malicious activity.

q) **Volatile Artifacts**

Data stored in RAM such as running processes, network connections, injected code, and command execution traces.

23. References

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